

Uniqueness of the Stationary Equilibrium in Aiyagari (1994): A Machine-Checked Proof for the Log Calibration, and the Frontier for Higher Risk Aversion

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Abstract

Aiyagari (1994) computed the stationary equilibrium of his heterogeneous-agent economy numerically. We prove that at his logarithmic calibration ($\mu = 1$) it is: for any of his labour-endowment processes, and for any finite Markov earnings process with monotone transitions, at most one interest rate in $(-\delta, \lambda)$ clears the capital market, where $\lambda = 1/\beta - 1$ is the rate of time preference; no equilibrium exists above λ or below -4% ; and the bisection he used is certified to converge to the rate. Every step is checked by the Lean 4 proof assistant against the Mathlib library. The economics is Light's (2020) chain, saving rises with the rate, so aggregate capital does, so supply meets a decreasing demand once, but at $\beta = 0.96$ the published constants fail; what succeeds is the Euler inequality at the borrowing constraint, a minimal marginal propensity to consume, a Doeblin atom, and homogeneity of the stationary distribution in the wage. For $\mu \in \{3, 5\}$ we locate the obstruction exactly: a condition on the response of consumption to the rate that fails for the rich. A tighter argument weakens it to one that holds numerically at every wealth level with an explicit margin, but that remains unproved. For Huggett (1993) the obstruction is the debtors, and we bound their effect.

JEL: C63, D52, E21. *Keywords*: Bewley models, uniqueness, stationary equilibrium, formal verification, Lean.

1 Introduction

The stationary equilibrium of Aiyagari (1994) is a number: the interest rate at which the capital that a continuum of households accumulates against uninsurable earnings risk equals the capital a competitive firm demands. Aiyagari computed it by bisection on the rate and reported it, for three degrees of relative risk aversion $\mu \in \{1, 3, 5\}$ and eight earnings processes,

*The Lean development described here is at <https://github.com/LeanEconomics/LeanEconomics>. It was written mostly by Claude (Anthropic), an AI coding model, working under the direction of Robert Kirkby (Victoria University of Wellington); every result reported was checked by the Lean kernel. Comments welcome.

in his Table II. He did not ask whether the number was unique, and for three decades the question was treated as settled by the picture: capital supply is drawn rising in the rate, demand falling, and the curves cross once.

The picture is not a theorem. Açıkgöz (2018) proved existence and gave conditions for uniqueness that do not cover Aiyagari’s parameters; Light (2020) proved uniqueness for relative risk aversion at most one, again under conditions on the discount factor and the earnings process that his own statement does not verify at $\beta = 0.96$; Light and Weintraub (2022) put the argument in a general mean-field framework; Kirkby (2019) computed the Aiyagari equilibrium with a discretisation proved to converge to the true solution, found numerically that it is almost certainly unique, and recorded that uniqueness remained open theoretically; Walsh and Young (2025) showed by computation that at high risk aversion the supply curve can bend backward and the equilibrium can fail to be unique. The survey of Toda and Walsh (2024) lists uniqueness in the Aiyagari model with its actual calibration as open.

This paper closes it for $\mu = 1$ and locates the obstruction for $\mu \in \{3, 5\}$, and it does so in a form that is new to economics: every result is checked by a proof assistant. The whole development, from the definition of the household problem to the theorem that two equilibrium rates of Aiyagari’s log economy coincide, is written in Lean 4 (de Moura and Ullrich, 2021) against the Mathlib library (The mathlib Community, 2020), and the Lean kernel has verified that each proof follows from its hypotheses and the standard axioms of the library. There are no unproved steps, no appeals to results that are “well known”, and no numerical approximations. The formal statement of each result reported below is named in the text, and Appendix A lists them with the files that contain them.

The main result. Fix Aiyagari’s preferences and technology at $\mu = 1$: log utility, $\beta = 0.96$, Cobb–Douglas production with capital share $\alpha = 0.36$ and depreciation $\delta = 0.08$, no borrowing. Let the labour endowment follow any finite-state Markov chain whose transitions are monotone (a higher state’s row first-order stochastically dominates a lower state’s) and whose lowest state is reached from every state in one step. Then at most one interest rate in $(-\delta, \lambda)$, with $\lambda = 1/\beta - 1$, is a stationary equilibrium rate (Theorem 3.1). Aiyagari’s own earnings processes, Tauchen discretisations of an AR(1) with positive persistence on any number of grid points, satisfy the two conditions by construction (Corollary 3.2). Outside $(-\delta, \lambda)$ there is nothing: above λ the household is patient and accumulates without bound, so no stationary distribution has finite mean capital below demand (Theorem 3.4); below $-\delta$ the only capital a log household holds is bounded by $\beta\bar{y}/(1 - \beta(1 + r))$, which falls under demand as demand explodes at $-\delta$ (Theorem 3.5). Existence on any interval is reduced to a single inequality on capital supply at its upper end (Theorem 3.6), and the bisection Aiyagari ran is proved to bracket and converge to the unique rate (Theorem 3.7).

Why the published arguments do not apply, and what does. The economics of the proof is the chain of Light (2020): individual saving rises with the rate (his Theorem 1); therefore aggregate capital, the mean of the stationary distribution, rises with the rate

(his Theorem 2); therefore supply meets the strictly decreasing demand at most once (his Theorem 3). Each link needed something new at $\beta = 0.96$.

Theorem 1 is where relative risk aversion enters, through the monotonicity of $cu'(c)$ that the rescaling argument needs, and for log utility that is available. But the argument is an induction along value-function iterates, and it needs the borrowing constraint to be slack at the states it compares and the asset cap, an artefact of compactness, to be slack everywhere. The published sufficient conditions for both involve Lipschitz constants of the value function that scale like $1/(1 - \beta R)$, which at Aiyagari's rates is several hundred; at his parameters the tests fail by orders of magnitude. What replaces them are two consequences of the Euler inequality itself: a household at the constraint consumes at least the lowest income, which turns the corner condition into the primitive inequality $\beta R u'(y_{\min}) < u'(\text{cash on hand})$ (Section 4.2); and a minimal marginal propensity to consume, $\kappa = 1 - \beta$ for log utility at every rate, which bounds saving above and makes the cap slack under a single inequality $2\beta\bar{y} < (1 - \beta R)\bar{a}$ (Section 4.3).

Theorem 2 is a coupling: the two economies are run from a common initial distribution, the higher rate's policy dominates pointwise, the Markov operator preserves the order, and the order passes to the limit. The limit exists and is the stationary distribution because of a Doeblin condition that the borrowing constraint supplies, an atom at zero assets after a run of bad shocks; the same atom gives uniqueness of the stationary distribution at each rate, which is what turns capital supply into a function of the rate (Section 4.5).

Theorem 3 is elementary once supply is a monotone function, but Aiyagari's equilibrium condition is stated at the equilibrium wage, and the household problem is solved at wage one. The bridge is homogeneity of the stationary distribution in the wage, proved at the level of measures rather than policies (Section 4.6), so that the condition $K(r) = \mathbb{E}_r[a]$ at wage $w(r)$ becomes $k(r)/w(r) = \mathbb{E}_r^1[a]$ for the unit-wage economy, and $k/w = \alpha/((1 - \alpha)(r + \delta))$ needs no root.

A last feature deserves mention because it shaped the whole development. Nothing is differentiated. The value function of a constrained household is not known to be differentiable where the constraint binds, and the standard envelope theorems do not apply there. The Euler inequalities that drive every argument are derived from one-sided perturbations of the optimal plan and secant slopes, in the manner of Clausen and Strub (2020); Lipschitz bounds are secant bounds; monotonicity is proved by comparing two plans, never by a derivative. This is not a stylistic choice: it is what allowed the proofs to be completed at all in a setting where a proof assistant will not let a differentiability assumption pass unnoticed.

Higher risk aversion. For $\mu \in \{3, 5\}$ the argument fails at Theorem 1, and we show exactly how. The Euler-chain proof of Theorem 1 reduces to a single *elasticity condition* on the consumption function (Section 5.1): at tomorrow's assets b , $R_1 u'(c_1(b)) \leq R_2 u'(c_2(b))$, that is, for CRRA, $(c_2/c_1)^\mu \leq R_2/R_1$. For $\mu > 1$ this is false at high wealth, by the asymptotic linearity of the consumption function (Ma and Toda, 2022): the asymptotic marginal propensity to consume rises with the rate when $\mu > 1$, so consumption of the very rich rises with the rate by more than the condition allows. At Aiyagari's numbers the

condition holds for households with wealth up to about thirteen times mean income at $\mu = 3$ and seven times at $\mu = 5$, and fails beyond. Saving of the rich nonetheless rises with the rate; the sufficient condition simply does not see it.

The chain, it turns out, throws a term away. It starts from the supposition that saving falls with the rate, which gives $c_2(a) > c_1(a) + (r_2 - r_1)a$, and uses only $c_2(a) > c_1(a)$. Keeping the extra interest income weakens the condition to $(c_2(b)/c_1(b))^\mu \leq (R_2/R_1)(1 + (r_2 - r_1)a/c_1(a))^\mu$ (Section 5.3), and in the high-wealth limit this holds for every μ , exactly, with a margin in logarithmic terms of $1/(\kappa R)$ per unit of rate difference, about 25 at Aiyagari's parameters. The slack condition is therefore numerically true at every wealth level. What we cannot do is prove it from primitives: it asks for the ratio of consumption at two rates to a relative precision of the order of the rate difference, uniformly in wealth, and no bound we have on the consumption function is that sharp. That is the frontier, and we state it as such.

Huggett. In Huggett (1993) the asset is a bond in zero net supply and households borrow. Against a constant supply, monotone capital supply is not enough for single crossing, strict monotonicity is needed, and we prove the strict form of Theorem 1 for creditors. For debtors a higher rate lowers cash on hand, and saving can fall with the rate by up to the extra interest on the debt. We carry this loss through the coupling and obtain $K_1 \leq K_2 + (r_2 - r_1)|\underline{a}|/(1 - \mathfrak{p}_2)$, where $\mathfrak{p}_2 = (\beta R_2)^{1/\sigma}$ is the marginal propensity to save of the rich (Section 6). At Huggett's numbers the amplification is about 1500, and we have no lower bound on the creditors' gain to set against it. We also show that the propensity-to-save bound the argument needs cannot be obtained on a bounded asset region from the concavity of the consumption function and its minimal-MPC floor, although it can on a ray, and we give the counterexample as a theorem.

What is and is not claimed. The result is about Aiyagari's model as he wrote it, with one difference: assets are confined to $[0, \bar{a}]$ for some cap $\bar{a}$, which his computation also imposes, and the theorem carries one inequality saying the cap is large enough not to bind. Everything else is his: the utility, the technology, the earnings process up to the two structural conditions. The uniqueness is of the equilibrium interest rate; the stationary distribution at that rate is unique as well, by the Doeblin argument. Existence is proved on any interval $[r_{lo}, r_{hi}] \subset (-\delta, \lambda)$ whose upper end carries enough capital supply to exceed demand; that supply floor is the one quantitative fact about the log household this paper does not prove, and Section 3.3 says why. The bisection theorem takes an equilibrium as given and certifies the algorithm.

Related work. Uniqueness in Bewley economies is treated by Açıkgöz (2018), Light (2020), Light and Weintraub (2022), and surveyed in Toda and Walsh (2024); multiplicity at high risk aversion is documented by Walsh and Young (2025). The consumption-function theory we lean on is Carroll and Kimball (1996) for concavity, Ma and Toda (2022) for asymptotic linearity and the minimal MPC, Carroll and Shanker (2026) for the buffer-stock bounds, and Li and Stachurski (2014) for the contraction in marginal-utility space that gives sufficiency of the Euler equation. The formalisation uses the fixed-point and monotone-operator results of

Sargent and Stachurski (2025), several of which we formalised along the way (Section 7). On formal verification in economics we know of no prior formalisation of a recursive macroeconomic equilibrium; the closest work is the mean-field-games literature on uniqueness, Lasry and Lions (2007), Graber and Matter (2024), Cannarsa and Capuani (2018), whose monotonicity conditions we discuss in Section 7 and show do not reach stationary discounted equilibria.

Section 2 sets out the model as formalised. Section 3 states the results for $\mu = 1$. Section 4 explains the proof. Sections 5 and 6 give the frontier for $\mu > 1$ and for Huggett. Section 7 collects auxiliary results. Appendix A is the theorem index and Appendix B describes the formalisation.

2 The model, as formalised

2.1 The household

A household economy is a tuple

$$P = (Z, y, \Pi, r, \beta, u, \underline{a}, \bar{a}).$$

Z is a finite set of income states; $y : Z \rightarrow \mathbb{R}$ is the labour endowment with $0 < y_{\min} \leq y(z) \leq y_{\max}$; Π is a stochastic matrix on Z ; $r > -1$ is the interest rate; $\beta \in [0, 1)$ is the discount factor; u is period utility; and $[\underline{a}, \bar{a}]$ with $\underline{a} \leq \bar{a}$, $0 \leq \bar{a}$, is the asset region. The borrowing limit and the cap are *type parameters* of the formal object, so that Aiyagari's economy ($\underline{a} = 0$) and Huggett's ($\underline{a} < 0$) are the same structure, and a theorem proved for a general floor applies to both. The state is $(a, z) \in [\underline{a}, \bar{a}] \times Z$, cash on hand is $m(a, z) = y(z) + (1 + r)a$, and the household solves

$$V(a, z) = \max_{a' \in [\underline{a}, \min\{\bar{a}, m(a, z)\}]} \left\{ u(m(a, z) - a') + \beta \sum_{z'} \Pi(z, z') V(a', z') \right\}. \quad (1)$$

Utility is required to be continuous, increasing and strictly concave on a domain that is either $(0, \infty)$ (utility unbounded below, as for log and CRRA with $\mu \geq 1$) or $[0, \infty)$; off the domain the reward is $-\infty$. With a negative borrowing limit the model carries Aiyagari's natural-borrowing-limit condition, $y_{\min} + r\underline{a} > 0$: income at the limit covers the interest on the debt. Because raising r raises the debt service, a rate is *admissible* for P , written $\text{RateOK}(r)$, when $1 + r > 0$ and the limit remains sustainable at r ; with $\underline{a} = 0$ the second condition is vacuous. $P.\text{withRate}(r)$ denotes the same economy at rate r .

The value function exists and is the unique bounded fixed point of (1); the policy $a' = g(a, z)$ is unique by strict concavity; consumption is $c(a, z) = m(a, z) - g(a, z)$. These are theorems of the development (a stochastic principle of optimality with extended-real rewards), not assumptions. Where a result needs consumption to be positive at the optimum it takes the hypothesis *PositiveConsumption*, which is discharged automatically when utility is unbounded below and by a marginal Inada condition otherwise.

2.2 Distributions

The state space $S = [\underline{a}, \bar{a}] \times Z$ is compact. The policy and the transition matrix define a Markov kernel on S , and its action on Borel probability measures is the operator $\mu \mapsto T\mu$. A *stationary distribution* is a fixed point, $T\mu = \mu$. Aggregate capital is mean assets,

$$K(\mu) = \int a d\mu(a, z).$$

Existence of a stationary distribution is by the Krylov–Bogolyubov argument on the compact state space, a theorem of the development; uniqueness is not automatic and is discussed in Section 4.5.

Two structural conditions on the earnings process recur. The transitions are *monotone* when for any z_1, z_2 with $y(z_1) \leq y(z_2)$, the row $\Pi(z_2, \cdot)$ first-order stochastically dominates $\Pi(z_1, \cdot)$ in the income order; iid income is the trivial case, and a Tauchen discretisation of an AR(1) with $\rho \geq 0$ is monotone by construction. A state z_0 is *reached from everywhere* when $\Pi(z, z_0) > 0$ for every z ; for the Doeblin argument z_0 is a state of minimal income.

2.3 The firm and the equilibrium condition

Aiyagari’s firm is Cobb–Douglas with capital share α and depreciation δ , with labour supplied inelastically and the endowment process normalised to mean one, so that per-capita capital demand and the wage at rate r are

$$k(r) = \left(\frac{\alpha}{r + \delta} \right)^{1/(1-\alpha)}, \quad w(r) = (1 - \alpha)k(r)^\alpha, \quad \frac{k(r)}{w(r)} = \frac{\alpha}{(1 - \alpha)(r + \delta)}.$$

His stationary equilibrium condition is $k(r) = \mathbb{E}_{r, w(r)}[a]$: capital demand equals mean asset holdings under the stationary distribution of the household economy at rate r and wage $w(r)$. The household problem at wage w is the problem at wage one with incomes and the cap scaled by w , and the stationary distribution scales with it, so the condition is equivalent to

$$\mathbb{E}_r^1[a] = D(r), \quad D(r) := \frac{k(r)}{w(r)} = \frac{\alpha}{(1 - \alpha)(r + \delta)}, \quad (2)$$

where $\mathbb{E}_r^1$ is taken under a stationary distribution of the unit-wage economy at rate r . The formal object `normalisedDemand( $\alpha, \delta, r$ )` is $D(r)$, and an *equilibrium rate* is an admissible r together with a stationary distribution μ of $P.\text{withRate}(r)$ satisfying $K(\mu) = D(r)$. The reduction from Aiyagari’s form to (2) is Section 4.6.

D is strictly decreasing on any interval with $r + \delta > 0$ and unbounded as $r \downarrow -\delta$. The rate of time preference is $\lambda = 1/\beta - 1$; at $\beta = 0.96$ it is $1/24 \approx 4.17\%$, and Aiyagari’s reported equilibrium rates at $\mu = 1$ (his Table II) lie between 3.31% and 4.17%; at $\mu \in \{3, 5\}$ they range down to -0.35% .

2.4 Aiyagari's calibration

Throughout, “Aiyagari's numbers” means $\beta = 24/25$, $\alpha = 9/25$, $\delta = 2/25$, $\underline{a} = 0$, log utility for $\mu = 1$ and $u(c) = c^{1-\mu}/(1-\mu)$ for $\mu \in \{3, 5\}$, and a labour endowment $\log l' = \rho \log l + \sigma \sqrt{1-\rho^2} \varepsilon$ with $\rho \in \{0, 0.3, 0.6, 0.9\}$, $\sigma \in \{0.2, 0.4\}$, discretised by Tauchen's method on $[-3\sigma, 3\sigma]$. The formal statements use the rational values; 0.96 is 24/25 exactly.

3 Results for the log calibration

Throughout this section P is a household economy with $\underline{a} = 0$, cap $\bar{a}$, log utility, $\beta = 24/25$, and an earnings process (Z, y, Π) . Two conditions on the earnings process are assumed in every uniqueness statement:

(E1) the transitions are monotone;

(E2) there is a state z_0 with $y(z_0) = y_{\min}$ and $\Pi(z, z_0) > 0$ for every z .

The cap enters through one inequality, evaluated at the larger of the two rates compared:

$$2\beta y_{\max} < (1 - \beta(1 + r)) \bar{a}. \quad (3)$$

It says the cap was set large enough not to bind; nothing else about $\bar{a}$ is used.

3.1 Uniqueness

Theorem 3.1 (Aiyagari at $\mu = 1$: the equilibrium rate is unique on $(-\delta, \lambda)$). *Let $r_1, r_2 \in (-2/25, 1/24)$ be admissible, and suppose that for $i = 1, 2$ there is a stationary distribution μ_i of P with $\text{Rate}(r_i)$ with $K(\mu_i) = D(r_i)$, where $D(r) = \alpha/((1-\alpha)(r+\delta))$ with $\alpha = 9/25$, $\delta = 2/25$. If (3) holds at $\max\{r_1, r_2\}$, then $r_1 = r_2$.*

Formal statement: *IncomeFluctuation.aiyagari1994_log_equilibriumRate_unique_100.*

The theorem is stated for every interval $[r_{lo}, r_{hi}]$ with $-\delta < r_{lo} \leq r_{hi} < \lambda$ in the form *aiyagari1994_log_equilibriumRate_unique*, with (3) at r_{hi} ; the open-interval version applies it to $[\min\{r_1, r_2\}, \max\{r_1, r_2\}]$. The lower end $-\delta$ is where demand becomes infinite; the upper end λ is where the household becomes patient. Both are excluded for a reason (Theorems 3.4 and 3.5), so Theorem 3.1 covers every rate at which an equilibrium could exist, up to a band above λ discussed below.

Corollary 3.2 (Aiyagari's own earnings processes). *Let $n \geq 0$, $\rho \in [0, 1)$, $\sigma > 0$, $c > 0$, and let the endowment take the $n+2$ values $c e^{-3\sigma+i\cdot 6\sigma/(n+1)}$, $i = 0, \dots, n+1$, with the Tauchen transition matrix of $\log l' = \rho \log l + \sigma \sqrt{1-\rho^2} \varepsilon$ on that grid. Then the conclusion of Theorem 3.1 holds, with (3) read with $y_{\max} = c e^{3\sigma}$.*

Formal statement: *aiyagariTauchen_equilibriumRate_unique_100.*

The corollary covers all eight of Aiyagari's processes and any finer grid. Its proof is that the Tauchen chain satisfies (E1) and (E2): the cumulative rows $F_i(k) = \Phi((x_k + h/2 - \rho x_i)/\sigma_\varepsilon)$ are decreasing in i because Φ is monotone and $\rho \geq 0$, which is first-order dominance, and

every entry of the first column is a value of Φ on the real line, hence positive. The normal distribution is Mathlib's `gaussianReal`, so Φ is a genuine cumulative distribution function and the positivity is a theorem about the Lebesgue integral, not a numerical fact. The scale c is whatever normalises the mean to one; uniqueness does not depend on it except through (3).

Remark 3.3 (What is used about the earnings process). Only (E1) and (E2). No transition probability is named, no stationary distribution of the chain is computed, and the number of states is arbitrary. (E1) is what Light's Theorem 1 needs to propagate monotonicity of the value function in income through the expectation; (E2) is what the Doeblin argument needs to produce an atom. For $\rho = 0$ both are immediate; for $\rho > 0$ they are Huggett's Lemma 1 for Tauchen chains.

3.2 No equilibrium outside $(-\delta, \lambda)$

Theorem 3.4 (Above the rate of time preference). *Let $r \geq 0$ with $\beta(1+r) > 1$, and let μ be stationary for $P.withRate(r)$. If*

$$y_{\max} + (1+r)D < \left(1 - \frac{1}{\beta(1+r)}\right)(y_{\min} + r\bar{a}),$$

then $K(\mu) \neq D$. In particular no equilibrium exists at r once $r - \lambda$ exceeds a quantity of order $(y_{\max} + (1+r)D(r))/(\beta r \bar{a})$.

Formal statement: `IncomeFluctuation.log_no_equilibrium_of_patient_risk`.

The proof is a supermartingale bound. With $\beta R > 1$ the Euler inequality makes $u'(c_t)$ a supermartingale that cannot decrease on average, so consumption cannot rise on average, while cash on hand grows at rate r on the part of the distribution near the cap; integrating against the stationary distribution gives a floor on $K(\mu)$ of order $(1 - 1/\beta R)r\bar{a}$. The band $(\lambda, \lambda + \epsilon)$ that escapes has width shrinking like $1/\bar{a}$, and with $\bar{a}$ chosen to satisfy (3) it is thin. A version for CRRA with integer risk aversion n , via Jensen on x^{-n} , is `crra_no_equilibrium_of_patient_mean`.

Theorem 3.5 (Negative rates). *Assume (E2) and that the earnings process is normalised so that the invariant distribution of Π has mean at most one. Let μ be stationary for $P.withRate(r)$.*

(i) *For any period utility satisfying the standing assumptions, if $r \in (-2/25, -13/250]$ then $K(\mu) \neq D(r)$.*

(ii) *For log utility, if $24y_{\max} < \bar{a}$ and $r \in (-2/25, -1/25]$ then $K(\mu) \neq D(r)$.*

Formal statement: `IncomeFluctuation.aiyagari1994_no_equilibrium_of_budget`;
`IncomeFluctuation.aiyagari1994_log_no_equilibrium_below_mean`.

Part (i) uses nothing about preferences: at a negative rate, integrating the budget constraint against a stationary distribution gives $K \leq \bar{Y}/(-r)$, the capital whose interest cost exhausts mean income, which is the zero-consumption supply schedule of Walsh and Young (2025). Part (ii) uses the log ceiling $K \leq \beta \bar{Y}/(1 - \beta(1+r))$ from the minimal MPC

(Section 4.3). Both need the income marginal of the stationary distribution to be the chain's invariant distribution, which is where (E2) enters, through a finite-chain uniqueness argument (`invariant_eq_of_reach`). In both cases the ceiling falls below $D(r)$, which grows without bound as $r \downarrow -\delta$.

Together, Theorems 3.1–3.5: for Aiyagari's log calibration the set of equilibrium rates is contained in $(-1/25, \lambda + \epsilon)$ and has at most one point in $(-\delta, \lambda)$.

3.3 Existence

Theorem 3.6 (Existence on an interval, given a supply floor at its top). *Let $-\delta < r_{lo} < r_{hi} < \lambda$, assume (E1), (E2) and (3) at r_{hi} . Suppose*

$$\frac{\beta y_{\max}}{1 - \beta(1 + r_{lo})} \leq D(r_{lo}), \quad D(r_{hi}) \leq m \leq K(\mu) \text{ for every stationary } \mu \text{ at } r_{hi}.$$

Then there is an equilibrium rate in $[r_{lo}, r_{hi}]$.

Formal statement: `IncomeFluctuation.aiyagari1994_log_exists_equilibrium_of_floor`.

The first inequality is the log capital ceiling at r_{lo} against demand there, and it holds for r_{lo} near $-\delta$ because demand is unbounded. The second is a floor m on capital supply at r_{hi} that exceeds demand there, and it is the one quantitative fact about the log household that this development does not prove. Aiyagari's equilibria sit where $k/w \approx 5$; the analytic supply floors available (an atom at the cap, positive saving from the corner) are nowhere near that, and how much an impatient household accumulates before impatience takes over is a quantitative question about the policy that the qualitative theory does not answer. With the floor supplied, existence follows by the intermediate value theorem, because capital supply along the unique stationary distribution is continuous in the rate (Section 4.7).

3.4 Computation

Theorem 3.7 (Bisection converges to the equilibrium rate). *Under the hypotheses of Theorem 3.1 on $[r_{lo}, r_{hi}]$, let ν be any selection of stationary distributions, $\nu(r)$ stationary for $P.\text{withRate}(r)$ on $[r_{lo}, r_{hi}]$, and let $E(r) = K(\nu(r)) - D(r)$ be excess supply. If $r^* \in [r_{lo}, r_{hi}]$ is an equilibrium rate, then the bisection on E started from $[r_{lo}, r_{hi}]$ brackets r^* at every step, its midpoint after n steps is within $(r_{hi} - r_{lo})/2^{n+1}$ of r^* , and its endpoints converge to r^* .*

Formal statement: `IncomeFluctuation.aiyagari1994_log_bisection`.

Excess supply is strictly increasing under these hypotheses, because supply is monotone (Section 4.4) and demand strictly decreasing, and strict monotonicity alone decides which half of a bracket contains the zero; neither continuity nor a sign condition at the ends is used. Any stationary distribution at r is the selection's, by the Doeblin uniqueness, so the algorithm does not depend on how the stationary distribution is computed. The theorem certifies the outer loop of the standard procedure, as in Kirkby (2017b), given exact inner solutions; convergence of the discretised inner solutions to the true ones is the subject of Kirkby (2017a) and Kirkby (2019), and is not re-proved here.

4 How the proof goes

The argument is Light's chain. This section describes each link and what had to be done to make it hold at $\beta = 0.96$. Formal names are given at each step; the emphasis is on the economics.

4.1 Theorem 1: saving rises with the rate

Fix two admissible rates $r_1 < r_2$ and write g_i for the policy at r_i . Light's Theorem 1 is $g_1(a, z) \leq g_2(a, z)$ for every state. His proof compares the two Bellman operators along value-function iterates: the map $v \mapsto v(ta, \cdot)$, $t = R_1/R_2$, turns a household at the higher rate with assets ta into one with the same cash on hand as a household at the lower rate with assets a , and the induction shows that the difference of the two value functions has increasing differences in assets, which is what makes the higher-rate policy larger. The step where the utility function enters is an inequality of the form $tu'(c(ta)) \leq u'(c(a))$ for the rescaled consumption, and this is where relative risk aversion at most one is used: it is monotonicity of $cu'(c)$, which for log utility is an identity. (`monotoneOn_bellman_sub_scaled_of_piv, policy_mono_withRate_ccrra.`)

The induction needs the constraint slack at the compared states and the cap slack everywhere along the iterates, and the published tests for both scale like $1/(1 - \beta R)$. At Aiyagari's rates $1 - \beta R$ is about 0.002, and the tests fail by orders of magnitude. What replaces them is in the next two subsections. With those in hand the induction is run locally, on an interval of rates short enough that the constants survive, and the local statements are chained across $[r_{lo}, r_{hi}]$ (`LargeBetaLight`).

4.2 The corner from the Euler inequality

Lemma 4.1 (Nobody at the constraint consumes less than the lowest income). *At an interior choice $g(a, z) > 0$ the Euler inequality reads $u'(c) \leq \beta R \mathbb{E}[u'(c')]$. Take the income state at which consumption at zero assets is smallest. If that household saved, tomorrow's consumption would be at least today's minimum, so $u'(c) \leq \beta R u'(c)$, impossible with $\beta R < 1$. So it is at the corner, consumes its income, and consumption at the constraint is at least $y_{\min}$ in every state.*

Formal statement: `IncomeFluctuation.minIncome_le_consumptionFn_floor.`

The corner condition at a state (a, z) is then the contrapositive of the Euler inequality: if $\beta R u'(y_{\min}) < u'(m(a, z))$, saving is not optimal, because saving would give $u'(m(a, z)) \leq u'(c) \leq \beta R \mathbb{E}[u'(c')] \leq \beta R u'(y_{\min})$. This is a primitive inequality, checkable by hand at $\beta = 0.96$, and it is what `aiyagari1994_log_equilibriumRate_unique` evaluates internally with an explicit corner level $a_0 = \min\{\bar{a}, y_{\min}(1 - \beta R)/(2\beta R^2)\}$. (`EulerCorner.`)

4.3 The cap from the minimal marginal propensity to consume

Lemma 4.2 (Minimal MPC). *For CRRA utility with parameter γ at a zero borrowing limit, with $\beta R < 1$,*

$$c(a, z) \geq \kappa m(a, z), \quad \kappa = 1 - \frac{b}{R}, \quad b = (\beta R)^{1/\gamma}.$$

Formal statement: *IncomeFluctuation.crra_minMPC_mul_le_consumptionFn.*

b is Carroll's absolute patience factor and κ the MPC of a perfect-foresight consumer with no human wealth (Carroll and Shanker, 2026). The bound is self-reproducing under the Bellman operator: if the continuation's consumption satisfies it, then at an interior choice the Euler inequality gives $c^{-\gamma} \leq \beta R (\kappa R a')^{-\gamma}$, hence $c \geq (\kappa R/b) a'$, and $(\kappa R/b)/(1 + \kappa R/b) = \kappa$ exactly because $1 - \kappa = b/R$; at the corner the household consumes everything. For log utility $\kappa = 1 - \beta$ at every rate, which is why the log results carry no rate-dependent constant.

Two things follow. Saving is at most $(1 - \kappa)m(a, z)$, so at the cap $g(\bar{a}, z) \leq \beta(y_{\max} + R\bar{a}) < \bar{a}$ whenever (3) holds, and the cap is slack along every iterate and at the fixed point (*crra_policy_lt_cap_of_minMPC*). And integrating against a stationary distribution, $K \leq \beta(\bar{Y} + RK)$, so

$$K(\mu) \leq \frac{\beta \bar{Y}}{1 - \beta R}, \tag{4}$$

the log capital ceiling used in Theorems 3.5 and 3.6 (*log_aggregateCapital_le, log_aggregateCapital_le_mean*).

4.4 Theorem 2: the coupling

Proposition 4.3 (Aggregate capital rises with the rate). *Let $g_1 \leq g_2$ pointwise on the region, and let the two Markov operators be run from a common initial distribution μ_0 , converging to stationary distributions μ_1 and μ_2 . Then μ_2 dominates μ_1 in the asset coordinate, and $K(\mu_1) \leq K(\mu_2)$.*

Formal statement: *IncomeFluctuation.monotoneOn_capitalSupply_of_policy_mono.*

Dominance is tested against bounded continuous functions that are increasing in assets at each income state. Three facts carry the argument: the Markov operator preserves that test class, because the policy is increasing in assets (*policy_mono*); a pointwise-larger policy gives a dominating operator on that class; and dominance is closed under weak limits. The asset coordinate itself is in the test class, so the mean is ordered. This is the argument of Light and Weintraub (2022), Theorem 2, which asks for monotonicity in one coordinate only; monotonicity in the whole state fails here because the policy need not rise with the income shock.

4.5 Uniqueness and convergence of the stationary distribution

The coupling needs the iterates from μ_0 to converge, and the equilibrium condition needs capital supply to be a function of the rate, not a correspondence. Both come from a Doeblin condition. The asset transition is deterministic given the shock, so two households with

different assets never meet, and no minorisation against a spread-out measure holds. The borrowing constraint supplies an atom: if a run of N draws of the worst income state z_0 drives every household to exactly zero assets, then every N -step transition charges the single point $(0, z_0)$ with probability at least $\Pi_{\min}(\cdot, z_0)^N$, which is a Doeblin condition. The contraction is run on the function side: the oscillation of $T^N h$ is at most $(1 - \epsilon)$ times that of h , so two stationary distributions agree on every bounded continuous function, and the iterates from any start converge weakly to the stationary distribution. (`stationary_unique`, `tendsto_pushProb_iterate`.)

That a run of bad shocks reaches the constraint is the *decline* property: above some asset level the household dissaves in the worst state. For an impatient household this is Açıkgöz (2018), Proposition 4, and it is proved here from the corner condition and the minimal MPC (`ImpatientDecline`). (E2) is what makes the run have positive probability from every state. At each rate in $[r_{lo}, r_{hi}]$ the conclusion is a unique stationary distribution, reached from every start (`log_existsUnique_isStationary`).

4.6 Theorem 3: single crossing, and Aiyagari's form of the condition

With supply a monotone function of the rate on $[r_{lo}, r_{hi}]$ and D strictly decreasing, two equilibrium rates coincide: this is `eq_of_monotoneOn_of_strictAntiOn` and needs no continuity. The remaining step is to connect (2) to Aiyagari's condition at the equilibrium wage.

Proposition 4.4 (Homogeneity of the stationary distribution). *Scaling every income and the cap by $w > 0$ scales the policy by w . The map $(a, z) \mapsto (wa, z)$ therefore commutes with the Markov operator, so it carries stationary distributions of the unit-wage economy to stationary distributions of the wage- w economy, and aggregate capital scales by w .*

Formal statement: *IncomeFluctuation.HomotheticDistribution*.

This is Açıkgöz's Proposition 7 at the level of measures. It is what lets the theorems be stated in Aiyagari's own form, $K(r) = \mathbb{E}_r[a]$ at $w(r)$, while the household side is solved once at wage one and the firm side reduces to $k/w = \alpha/((1 - \alpha)(r + \delta))$, which needs no root (`aiyagari1994_equilibriumRate_unique_wage`).

4.7 Continuity of supply, for existence

Existence uses the intermediate value theorem, which needs capital supply continuous in the rate. The rate is not a parameter of the household problem in the formal development but part of its state, so that a family of problems indexed by the rate is one problem on a product space, and continuity of the policy in the rate is joint continuity of one policy. Weak continuity of the stationary distribution in the rate then follows from the Doeblin uniqueness (a limit of stationary distributions is stationary, and there is only one), and aggregation is weakly continuous because the asset coordinate is bounded and continuous (`continuousOn_capitalSupply`).

4.8 What is never differentiated

Every inequality above is derived without a derivative of the value or policy function. The Euler inequalities come from perturbing the optimal plan by ϵ in one direction and letting ϵ shrink: carrying the poorer household's plan forward, or the richer household's plan back, gives one-sided bounds on the marginal value of assets, and the first-order condition is read off the pair (`euler_le`, `euler_ge`; the construction is that of Clausen and Strub, 2020). Lipschitz constants are secant bounds; monotonicity in the rate is a comparison of two plans. Where the development does establish differentiability of the value function (`IncomeFluctuationEnvelope`) it uses the Clausen–Strub sandwich and not the Benveniste–Scheinkman argument, and the uniqueness proof does not depend on it.

5 Higher risk aversion: the frontier for $\mu \in \{3, 5\}$

Everything in Section 4 except Theorem 1 goes through for CRRA utility with any $\mu > 0$: the corner, the minimal MPC with $\kappa = 1 - \mathfrak{p}/R$, the cap slack, the Doeblin uniqueness and convergence, the coupling, the homogeneity, single crossing. The formal statement is `crra_equilibriumRate_unique_of_marginal`, and it carries exactly one hypothesis that is not discharged: Theorem 1. This section says what that hypothesis is, why it fails at Aiyagari's numbers, what it becomes when the proof is tightened, and where that leaves the problem.

5.1 Theorem 1 reduces to an elasticity condition

Proposition 5.1 (Saving rises with the rate under an elasticity condition). *Let $r_1 \leq r_2$ be admissible, with positive consumption and a slack cap at both rates, and let u' be strictly decreasing. Fix (a, z) with $a \geq 0$ and let $b = g_1(a, z)$. If*

$$R_1 u'(c_1(b, z')) \leq R_2 u'(c_2(b, z')) \quad \text{for every } z', \quad (5)$$

then $g_1(a, z) \leq g_2(a, z)$.

Formal statement: `IncomeFluctuation.policy_le_policy_withRate_of_marginal_at`.

The proof is four lines of Euler inequalities and holds for any period utility. Suppose $g_2(a, z) < g_1(a, z) = b$. Then the higher-rate household has room under the cap, so $\beta R_2 \mathbb{E}[u'(c_2(g_2, \cdot))] \leq u'(c_2(a, z))$; the lower-rate household has room above the floor, so $u'(c_1(a, z)) \leq \beta R_1 \mathbb{E}[u'(c_1(b, \cdot))]$; it consumes strictly less today, so $u'(c_2(a, z)) < u'(c_1(a, z))$; and consumption rises with assets, so $u'(c_2(b, \cdot)) \leq u'(c_2(g_2, \cdot))$. Chaining, $R_2 \mathbb{E}[u'(c_2(b, \cdot))] < R_1 \mathbb{E}[u'(c_1(b, \cdot))]$, which (5) forbids. For CRRA, (5) is

$$\left(\frac{c_2(b, z')}{c_1(b, z')} \right)^\mu \leq \frac{R_2}{R_1},$$

an elasticity of consumption to the gross rate of at most $1/\mu$ (`crra_elasticity_iff`). It says that the income effect of the rate on consumption does not outweigh the human-wealth effect. Uniqueness on $[r_{lo}, r_{hi}]$ follows from (5) at every state and every pair of rates (`crra_`

equilibriumRate_unique_of_marginal), and, since the coupling only visits states the economy visits, from (5) on any interval $[0, \bar{a}']$ that the lower-rate policy maps into itself (crra_equilibriumRate_unique_of_marginal_on, Section 5.5).

5.2 The condition fails for the rich when $\mu > 1$

By Ma and Toda (2022), the consumption function is asymptotically linear, $c(x)/x \rightarrow \kappa(R)$, with $\kappa(R) = 1 - (\beta R)^{1/\mu}/R$ the minimal MPC. For $\mu > 1$, κ rises with R . So at high wealth $c_2(b)/c_1(b) \rightarrow \kappa_2 R_2/(\kappa_1 R_1)$ and

$$\left(\frac{\kappa_2 R_2}{\kappa_1 R_1}\right)^\mu = \left(\frac{\kappa_2}{\kappa_1}\right)^\mu \left(\frac{R_2}{R_1}\right)^\mu > \frac{R_2}{R_1},$$

so (5) fails for all large b (crra_elasticity_fails_of_asymptotic_mpc). Saving of the rich nonetheless rises with the rate, since $g \approx \mathfrak{p}b$ with $\mathfrak{p} = (\beta R)^{1/\mu}$ increasing in R ; the sufficient condition does not see it. With the linear approximation $c \approx \kappa(R)(Rb + H)$ and $H \approx \bar{Y}/r$, the human-wealth effect dominates and consumption falls with the rate for b below about $13\bar{Y}$ at $\mu = 3$ and $7\bar{Y}$ at $\mu = 5$, where (5) holds; above, it fails. The stationary distribution reaches far beyond those levels, because the top income type keeps accumulating until wealth of order $y_{\max}/(1 - \mathfrak{p})$, so the invariant-interval form does not rescue the argument at Aiyagari's numbers.

5.3 The chain throws a term away

The supposition $g_2(a, z) < g_1(a, z)$ gives not $c_2(a, z) > c_1(a, z)$ but

$$c_2(a, z) > c_1(a, z) + (r_2 - r_1)a,$$

and the proof above uses only the weaker inequality. Keeping the extra interest income, the chain closes under a weaker condition.

Proposition 5.2 (The slack elasticity condition). *In the setting of Proposition 5.1, with $u' > 0$, if*

$$R_1 u'(c_1(b, z')) u'(c_1(a, z) + (r_2 - r_1)a) \leq R_2 u'(c_2(b, z')) u'(c_1(a, z)) \quad \text{for every } z', \quad (6)$$

then $g_1(a, z) \leq g_2(a, z)$. For CRRA, (6) reads

$$\left(\frac{c_2(b, z')}{c_1(b, z')}\right)^\mu \leq \frac{R_2}{R_1} \left(1 + \frac{(r_2 - r_1)a}{c_1(a, z)}\right)^\mu.$$

Formal statement: *IncomeFluctuation.policy_le_policy_withRate_of_slack_at; crra_slack_elasticity_iff.*

The new factor is at least one and grows with $a/c_1(a)$, which is largest exactly where (5) fails. In the high-wealth limit, with $L_i = \kappa_i R_i = R_i - \mathfrak{p}_i$ the asymptotic slopes and $b \approx \mathfrak{p}_1 a$, $c_1(a) \approx L_1 a$, both sides have limits and the inequality between them is exact:

Table 1: The slack elasticity condition at Aiyagari’s numbers, $r \approx 4\%$. Rows two and three are the logarithmic growth rates of the two sides of (6) per unit of rate difference in the high-wealth limit; the condition holds when the left is below the right. Row four is the wealth below which consumption falls with the rate, where the unweakened condition (5) already holds.

	$\mu = 3$	$\mu = 5$
minimal MPC κ	0.0390	0.0388
left side, deep tail	50	100
right side, deep tail	75	125
wealth below which (5) holds (multiples of mean income)	≈ 13	≈ 7

Proposition 5.3 (The slack condition holds in the homogeneous limit for every μ). *For $0 < R_1 \leq R_2$, $b_1 \leq b_2$, $b_1 < R_1$, $b_2 < R_2$ and $\mu \geq 0$,*

$$\left(\frac{R_2 - b_2}{R_1 - b_1}\right)^\mu \leq \frac{R_2}{R_1} \left(\frac{R_2 - b_1}{R_1 - b_1}\right)^\mu,$$

with strict inequality when $b_1 < b_2$ and $\mu > 0$. Consequently, along tomorrow’s assets, if $c_2(b(a))/c_1(b(a)) \rightarrow L_2/L_1$ and $c_1(a)/a \rightarrow L_1$, the CRRA slack condition holds for all large a .

Formal statement: `cr ra_slack_elasticity_of_patience; cr ra_slack_elasticity_holds_of_asymp totic_mpc`.

The proof is one line: $R_2 - b_2 \leq R_2 - b_1$ because $b_1 \leq b_2$, and $R_2/R_1 \geq 1$ is spare. In logarithmic terms per unit of rate difference the two sides grow at rates $\mu(\kappa'R + \kappa)/(\kappa R)$ and $1/R + \mu/(\kappa R)$, whose difference is exactly $1/(\kappa R)$ for every μ : about 25 at Aiyagari’s numbers. Table 1 gives the values.

The table is computed at $r \approx 4\%$, where Aiyagari’s equilibria at $\mu = 1$ and most at $\mu \in \{3, 5\}$ sit. His Table II also has equilibria at much lower rates, down to -0.35% at $\mu = 5$, $\sigma = 0.4$, $\rho = 0.9$, where κ is far smaller (0.005); the identity right – left = $1/(\kappa R)$ holds at any rate, so the tail margin there is larger, about 190.

So at every wealth level the slack condition is, numerically, true: in the body because consumption falls with the rate, in the tail by Proposition 5.3 with margin, and in between with margin as well. The reduction theorems with the slack condition in place of (5) are `equilibriumRate_unique_of_slack`, `equilibriumRate_unique_of_slack_on` and `cr ra_equilibriumRate_unique_of_slack_on`.

5.4 What remains

To prove (6) from primitives one needs the ratio $c_2(b)/c_1(b)$ of consumption at two rates to a relative precision of the order of $r_2 - r_1$, uniformly in b on the support, since that is the size of the margin. Two available routes do not deliver it. The sandwich $\kappa x \leq c(x) \leq \kappa x + C$, with C of the order of mean income, has a gap of order one and so pins the ratio only for rate pairs far enough apart, whereas uniqueness must handle all pairs. The fixed-point perturbation bound for the Coleman operator in the marginal-utility metric of Li and Stachurski (2014),

which we formalised (Section 7), gives a bound that is uniform in wealth in the wrong sense: it loses a factor of the inverse contraction modulus, $1/(1 - \mathfrak{p}) \approx 2000$ at these numbers, and is vacuous where u' is small. What would be needed is a Lipschitz estimate in the rate for the consumption function that is relative rather than absolute, that is, comparative statics of c in r uniform on the support. We know of no such result. The status of $\mu \in \{3, 5\}$ is therefore: reduced to an inequality on two consumption functions that is numerically true everywhere, with an explicit margin, and unproved.

Two further reductions are recorded for completeness. The rescaling proof of Section 4.1 uses relative risk aversion at most one only through a range condition on the true consumption function of the higher-rate economy, $(1 - c_2(0)/c_2(ta))(\rho - 1) \leq \rho^{1/\mu} - 1$ with $\rho = R_2/R_1$, roughly $c(a) \lesssim \frac{\mu}{\mu-1}c(0)$ (`policy_mono_withRate_crra_of_range`). It is weaker than $\mu \leq 1$ but fails at moderate wealth. And Theorem 1 holds at equal cash on hand for all a : $g_1(a, z) \leq g_2(ta, z)$ with $t = R_1/R_2$ (`policy_le_policy_scaled_withRate_of_marginal_at`).

5.5 The invariant-interval form

Since Theorem 1 uses the condition only at tomorrow's assets and the coupling only along coupled paths, if $[0, \bar{a}']$ is invariant under the lower-rate policy then the condition on $[0, \bar{a}']$ alone orders the policies there, the coupling started at zero assets never leaves the interval, and uniqueness follows (`AboveSet`, `push_aboveSet_eq_zero`, `equilibriumRate_unique_of_marginal_on`). This is the honest form of the reduction: the condition is needed on the ergodic support and nowhere else. At Aiyagari's numbers the support is too long for the unweakened condition, as noted; for the slack condition the invariant form is available but, by Proposition 5.3, not needed.

6 Huggett (1993)

In Huggett's exchange economy households trade a bond in zero net supply at price q ; the rate is $r = 1/q - 1$, the borrowing limit $\underline{a} < 0$ is a credit limit, and the equilibrium condition is $K(\mu) = 0$. His calibration is $\beta = 0.99322$, $\sigma \in \{1.5, 3\}$, a two-state endowment, and credit limits $\underline{a} \in \{-2, -4, -6, -8\}$ in units of the high endowment. Two things change.

Single crossing needs strictness. Against a constant supply, a monotone $K(r)$ gives nothing: two rates with $K = 0$ are compatible with $K \equiv 0$ between them. What is needed is that K be strictly increasing, and this follows if the higher-rate policy dominates the lower-rate one everywhere and strictly on a set the lower-rate stationary distribution charges (`aggregateCapital_lt_of_policy_lt`, `huggett_equilibriumRate_unique`). The strict form of Theorem 1 holds for creditors: at $a > 0$ with an interior choice, and $r_1 < r_2$, $g_1(a, z) < g_2(a, z)$ under (5) (`policy_lt_policy_withRate_of_marginal_at`).

Debtors lose cash on hand when the rate rises. For $a < 0$, the step $c_2(a) > c_1(a) + (r_2 - r_1)a$ has a negative last term, and the chain gives only

$$g_1(a, z) \leq g_2(a, z) + (r_2 - r_1) \max\{0, -a\} : \quad (7)$$

saving can fall with the rate for a debtor, but by no more than the extra interest on the debt (`policy_le_policy_add_withRate_of_marginal_at`; the slack version is `policy_le_policy_add_withRate_of_slack_at`). For $a \geq 0$ the loss vanishes and Theorem 1 is as before.

Carrying the loss through the coupling. The coupling of Section 4.4 needs exact dominance. A version that tolerates a shift replaces first-order dominance by *dominance up to δ* : $\int h d\mu \leq \int h d\nu + \delta L$ for every test function increasing in assets and L -Lipschitz in assets. One period of the operator turns a shift δ into $\delta K + \gamma$, where γ is the pointwise policy gap (7) and K is a Lipschitz constant of the higher-rate policy in assets; iterating and passing to the limit,

$$K(\mu_1) \leq K(\mu_2) + \frac{(r_2 - r_1) |a|}{1 - K} \quad \text{if } K < 1 \quad (8)$$

(`DominatesShift`, `dominatesShift_iterate`, `huggett_aggregateCapital_le_add`). Here K is the marginal propensity to save out of assets. For CRRA the policy's slope in assets is $R(1 - \text{MPC})$, and asymptotically the MPC is κ , giving $K = R(1 - \kappa) = \mathfrak{p} = (\beta R)^{1/\sigma} < 1$: impatience, with almost no room. At Huggett's numbers $1 - \mathfrak{p}_2 \approx 0.00067$, so the amplification in (8) is about 1500.

The propensity-to-save bound cannot be had on a bounded region. It is tempting to derive the MPC floor from Carroll–Kimball concavity of the consumption function together with the minimal-MPC level bound $c(x) \geq \kappa x$. On a ray this works: a concave function above the line κx has every secant slope at least κ , by one chord far to the right (`secant_ge_of_concaveOn Ici`, with the chord chosen explicitly). On the region $[a, \bar{a}]$ it does not: the constant function at the level of the bound at the cap is concave, lies above the bound, and is flat (`exists_concaveOn_ge_flat`). The MPC floor is an asymptotic property of the uncapped problem, and the chord at the cap says nothing because there the level bound is far below actual consumption. So (8) is restated with the floor as a hypothesis in the form it is used: a consumption slope floor $(R_2 - K)(b - a) \leq c_2(b) - c_2(a)$ is exactly the Lipschitz bound K for the policy (`policy_lipschitz_of_consumption_slope`), and the CRRA form takes $K = \mathfrak{p}_2$ and the floor $\kappa_2 R_2(b - a) \leq c_2(b) - c_2(a)$ (`ccrra_huggett_aggregateCapital_le_add_of_slack`).

Where Huggett stands. Uniqueness reduces to a quantitative race: the creditors' strict gain in `huggett_equilibriumRate_unique` against the debtors' loss bounded by (8). We have the sign of the gain and no lower bound on its size; every attempt at one runs into a pivot condition of the type (5). And the MPC floor on the region is the same asymptotic structure that blocks the slack condition at $\mu > 1$. Both are open.

7 Auxiliary results

The development contains a number of results that were built as tools for the main theorem or as tests of the frontier and stand on their own. This section lists the ones an economist may want.

Multiplicity is exactly non-monotone supply, and the equilibrium system permits it. Two distinct equilibrium rates force capital supply to be non-monotone on any interval containing them (`not_monotoneOn_of_equilibria_ne`), so every multiplicity example in the literature is a counterexample to Light’s Theorem 1 and to nothing else. Conversely an explicit continuous, strictly positive supply schedule, Cobb–Douglas demand plus a cubic wiggle, has exactly three equilibria with the demand $D(r) = A^2/(4(r + \delta)^2)$ (`wiggleSupply_equilibria`), matching the three-equilibrium shape Walsh and Young (2025) report: nothing in the firm side, the fixed-point machinery or the intermediate-value argument rules multiplicity out. What is not proved is that such a schedule is the capital supply of any household; the two bounds available, the gain test and the minimal MPC, both move the wrong way for a backward bend.

All capital is precautionary. A two-state economy near log utility holds at least $1/400$ of capital at every stationary distribution. The same economy with its income risk removed, both states replaced by their mean, saves nothing at any asset level, so its capital supply is exactly zero (`riskless_aggregateCapital_eq_zero`). The corner condition of Section 4.2 holds at every asset level in the riskless economy and fails above a threshold in the risky one, and the gap is the value of the insurance the household is buying.

Impatience is necessary. Marcet et al. (2007), Proposition 3: no stationary equilibrium with finite capital exists at $\beta R \geq 1$. Formalised with income risk through the Euler route, requiring $\beta R > 1$ strictly, and deterministically through secant slopes (`Impatience`); the version with a supply floor is Theorem 3.4.

Carroll–Kimball concavity across HARA. The consumption function is concave in assets for CRRA, shifted CRRA (via a power-mean inequality), and CARA (via a soft-minimum construction); quadratic utility is proved inadmissible because it is not increasing (`not_monotoneOn_quadUtility`). The induction runs along Bellman iterates, so it needs the Euler equation one step at a time and at states where the constraint binds, which is what `IncomeFluctuationEuler` supplies. A witness shows Carroll and Kimball’s conclusion is false when the cap binds (`not_concaveOn_consumptionFn_of_cap_binds`), which is why the cap slack condition cannot be dropped.

A calibrated class with existence and uniqueness. Before the log calibration was reached, existence and uniqueness were proved for a class of economies (`Calibrated`) each of whose numerical fields is one primitive inequality: near-log CRRA, log Stone–Geary

(`stoneGeary_exists_equilibrium_of_technology'`), and the $b \neq 0$ branch of HARA, with general finite Markov income. These carry the machinery of Section 4 at parameters where the published constants do work, and a CES witness proved uncalibratable shows the cap conditions are not vacuous.

Sufficiency of the Euler equation, without differentiability of the value function.

A solution of the functional Euler equation with equality at interior choices and the correct inequality at the constraint is the optimal consumption function (`eq_consumptionFn_of_isEulerSolution`). The proof is the contraction of the Coleman operator in the marginal-utility supremum metric of Li and Stachurski (2014), run on a class of candidate policies with a consumption floor and a saving slack. This is the sufficient direction of the Coleman–Reffett conjugacy of Sargent and Stachurski (2025), Proposition 8.3.13, proved without the conjugacy.

Abstract fixed-point results from Sargent and Stachurski. Parametric monotonicity of fixed points (Vol. 1, Prop. 2.2.7); eventually contracting maps (Thm. 6.1.5), which is what spectral-radius conditions deliver; the fitted-value-iteration error bound (Vol. 2, Thm. 9.1.3), the value-error input to the policy error bounds of Li (2015); and uniqueness of fixed points of increasing concave maps, in one dimension and in Du’s form on a finite product (`OrderedFixedPoints`). Santos (2000)’s policy error bound was assessed and not formalised: it needs interior solutions, which the borrowing constraint denies.

Mean-field-game uniqueness, and why it does not reach Bewley. The Lasry–Lions argument for uniqueness of mean-field equilibria adds two optimality inequalities. Its measure-theoretic core is formalised for the static game and for the finite-horizon finite-state game on paths: two mean-field Nash distributions have nonpositive Lasry–Lions pairing; under monotone costs the pairing vanishes; under strict monotonicity the costs coincide, though the distributions need not (`cost_eq_of_nash`, `value_eq_of_pathEquilibrium`; the static form of Cannarsa and Capuani, 2018, Thm. 4.1). The argument cancels $\int (u_1 - u_2) d(m_1(0) - m_2(0))$ by a common initial distribution, which a stationary discounted equilibrium does not have; and Graber and Matter (2024) show that games coupled through a market price are typically not Lasry–Lions monotone, the right condition there being monotonicity of the error map $Q \mapsto Q - \text{best response}(Q)$, which for Aiyagari is Light’s single crossing. The quantitative version, a strongly monotone Lipschitz operator on a real inner product space has a unique zero found by a damped iteration with modulus $\sqrt{1 - m^2/L^2}$, is formalised (`exists_unique_zero_of_stronglyMonotone`); for excess supply it would need a slope floor and a Lipschitz constant in the rate, neither of which is available.

Tauchen’s method. The Tauchen transition matrix is defined from cumulative rows of the standard normal distribution function, taken from Mathlib’s Gaussian measure, and proved to be stochastic, positive in its first column, and monotone in the sense of (E1) for $\rho \geq 0$ (`monotoneTransitions_of_tauschen`). This is what allows Corollary 3.2 to be stated for any number of grid points.

8 Conclusion

Aiyagari’s log economy has a unique stationary equilibrium, and the statement is now a theorem about his model as he wrote it, with his parameters, for any of his earnings processes, checked by a machine. The proof is the one the picture suggested, supply rises and demand falls, but the constants in the published versions of that argument do not survive $\beta = 0.96$, and what does survive are consequences of the Euler inequality at the borrowing constraint: a corner condition, a minimal marginal propensity to consume, and a Doeblin atom. None of it uses a derivative of the value function.

For $\mu \in \{3, 5\}$ we have reduced uniqueness to an inequality between two consumption functions that holds numerically at every wealth level with an explicit margin and that no available bound proves; for Huggett, to a quantitative race between creditors and debtors whose creditor side we cannot bound below. We regard both as genuinely open and have tried to say precisely what a proof would need: comparative statics of the consumption function in the interest rate, uniform on the support in relative terms. That is a question about the income fluctuation problem, not about equilibrium, and it may be the right next target.

On method, the experience of this development is that a proof assistant changes what one writes down. Arguments that quantify over “a sufficiently large cap” or “a small enough interval of rates” had to be given their constants, and at Aiyagari’s parameters several of the published constants were found to be too small by orders of magnitude; the replacements are simpler than the originals. Differentiability, which enters every textbook treatment through the envelope theorem, was never needed. And the exact location of the obstruction at $\mu > 1$, the wasted interest-income term in a four-line argument, was found by reading the formal proof rather than the informal one.

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A Theorem index

Each result stated in the text, with the Lean declaration that proves it and the file that contains it. All files are under `LeanEconomics/`. Every declaration listed depends only on the three standard axioms of `Mathlib`, `propxt`, `Classical.choice` and `Quot.sound`, as reported by `#print axioms`; the development contains no `sorry`.

Result	Declaration	File
Theorem 3.1	<code>aiyagari1994_log_equilibriumRate_unique_Ioo</code>	<code>Equilibrium/AiyagariUniqueness</code>
Theorem 3.1, interval form	<code>aiyagari1994_log_equilibriumRate_unique</code>	<code>Equilibrium/AiyagariUniqueness</code>
Theorem 3.1, at the wage	<code>aiyagari1994_equilibriumRate_unique_wage</code>	<code>Equilibrium/HomotheticDistribution</code>
Corollary 3.2	<code>aiyagariTauchen_equilibriumRate_unique_Ioo</code>	<code>Models/AiyagariTauchenWitness</code>
Tauchen chain monotone	<code>monotoneTransitions_of_tauschen</code>	<code>Models/Tauschen</code>
Theorem 3.4	<code>log_no_equilibrium_of_patient_risk</code>	<code>Equilibrium/RateRange</code>
Theorem 3.4, CRRA n	<code>crra_no_equilibrium_of_patient_mean</code>	<code>Equilibrium/Localisation</code>
Theorem 3.5(i)	<code>aiyagari1994_no_equilibrium_of_budget</code>	<code>Equilibrium/Localisation</code>
Theorem 3.5(ii)	<code>aiyagari1994_log_no_equilibrium_below_mean</code>	<code>Equilibrium/IncomeMarginal</code>
Theorem 3.6	<code>aiyagari1994_log_exists_equilibrium_of_floor</code>	<code>Equilibrium/AiyagariExistence</code>
Theorem 3.7	<code>aiyagari1994_log_bisection</code>	<code>Equilibrium/Bisection</code>
Theorem 1, log, local	<code>crra_policy_mono_withRate_of_minMPC</code>	<code>Models/LargeBetaLight</code>
Theorem 1, rescaling core	<code>monotoneOn_bellman_sub_scaled_of_piv</code>	<code>Models/IncomeFluctuationRateStep</code>
Corner from Euler	<code>minIncome_le_consumptionFn_floor</code>	<code>Models/EulerCorner</code>
Minimal MPC	<code>crra_minMPC_mul_le_consumptionFn</code>	<code>Models/MinimalMPC</code>
Cap slack	<code>crra_policy_lt_cap_of_minMPC</code>	<code>Models/MinimalMPC</code>
Log ceiling (4)	<code>log_aggregateCapital_le</code>	<code>Equilibrium/AiyagariUniqueness</code>
Theorem 2, coupling	<code>monotoneOn_capitalSupply_of_policy_mono</code>	<code>Equilibrium/CapitalSupplyMonotone</code>
Doebelin uniqueness	<code>stationary_unique</code>	<code>Distribution/Uniqueness</code>
Doebelin convergence	<code>tendsto_pushProb_iterate</code>	<code>Distribution/Uniqueness</code>

Result	Declaration	File
Unique stationary distribution, log	log_existsUnique_isStationary	Equilibrium/ AiyagariUniqueness
Theorem 3, single crossing	eq_of_monotoneOn_of_strictAntiOn	Equilibrium/Uniqueness
Homogeneity of the distribution	isStationary_scaleProb_log	Equilibrium/ HomotheticDistribution
Continuity of supply	continuousOn_capitalSupply	Equilibrium/ AiyagariContinuity
Euler inequalities	euler_le, euler_ge	Models/ IncomeFluctuationEuler
Income marginal is invariant	invariant_eq_of_reach	Equilibrium/IncomeMarginal
Proposition 5.1	policy_le_policy_withRate_of_marginal_at	Equilibrium/ ElasticityReduction
CRRA form of (5)	crra_elasticity_iff	Equilibrium/InvariantRegion
Uniqueness under (5)	crra_equilibriumRate_unique_of_marginal	Equilibrium/ ElasticityReduction
Invariant-interval form	crra_equilibriumRate_unique_of_marginal_on	Equilibrium/InvariantRegion
Failure at high wealth	crra_elasticity_fails_of_asymptotic_mpc	Equilibrium/InvariantRegion
Proposition 5.2	policy_le_policy_withRate_of_slack_at	Equilibrium/SlackEuler
CRRA form of (6)	crra_slack_elasticity_iff	Equilibrium/SlackEuler
Proposition 5.3	crra_slack_elasticity_of_patience	Equilibrium/SlackEuler
Proposition 5.3, eventual form	crra_slack_elasticity_holds_of_asymptotic_mpc	Equilibrium/SlackEuler
Uniqueness under (6)	crra_equilibriumRate_unique_of_slack_on	Equilibrium/SlackEuler
Range condition	policy_mono_withRate_crta_of_range	Models/ IncomeFluctuationRateStep
Equal cash on hand	policy_le_policy_scaled_withRate_of_marginal_at	Equilibrium/DebtorRate
Huggett single crossing	huggett_equilibriumRate_unique	Equilibrium/Huggett
Strict Theorem 1, creditors	policy_lt_policy_withRate_of_marginal_at	Equilibrium/DebtorRate
Debtor loss (7)	policy_le_policy_add_withRate_of_marginal_at	Equilibrium/DebtorRate
Shifted dominance (8)	huggett_aggregateCapital_le_add	Equilibrium/ShiftedDominance
Secant slope on a ray	secant_ge_of_concaveOn_Ici	Equilibrium/SavingPropensity
Bounded-region obstruction	exists_concaveOn_ge_flat	Equilibrium/SavingPropensity
MPC floor is a Lipschitz bound	policy_lipschitz_of_consumption_slope	Equilibrium/SavingPropensity
Huggett bound, CRRA, slack	crra_huggett_aggregateCapital_le_add_of_slack	Equilibrium/SavingPropensity

Result	Declaration	File
Multiplicity forces non-monotone supply	<code>not_monotoneOn_of_equilibria_ne</code>	Equilibrium/Multiplicity
Three equilibria	<code>wiggleSupply_equilibria</code>	Equilibrium/Multiplicity
Riskless twin saves nothing	<code>riskless_aggregateCapital_eq_zero</code>	Models/Precautionary
Euler sufficiency	<code>eq_consumptionFn_of_isEulerSolution</code>	Models/ColemanReffett
Sargent–Stachurski results	<code>fixedPoint_le_of_dominates, tendsto_iterate_of_eventuallyContracting, dist_fixedPoint_le_of_fittedVFI, eq_of_fixedPt_of_concave_du</code>	DynamicProgramming/OrderedFixedPoints
Lasry–Lions, static	<code>cost_eq_of_nash</code>	Equilibrium/LasryLions
Lasry–Lions, finite horizon	<code>value_eq_of_pathEquilibrium</code>	Equilibrium/LasryLions
Strongly monotone operators	<code>exists_unique_zero_of_stronglyMonotone</code>	Analysis/StronglyMonotone
Quadratic utility inadmissible	<code>not_monotoneOn_quadUtility</code>	Models/Quadratic
Concavity fails when the cap binds	<code>not_concaveOn_consumptionFn_of_cap_binds</code>	Models/IncomeFluctuationCarrollKimball

B The formalisation

What was checked. The development is written in Lean 4 (de Moura and Ullrich, 2021), toolchain v4.34.0-rc2, against the Mathlib library (The mathlib Community, 2020) at a fixed revision recorded in the repository manifest. It comprises 115 source files, about 35,000 lines, and about 1,500 theorem declarations. A Lean proof is a term whose type is the statement; the kernel checks that the term is well-formed. What the reader is asked to trust is the kernel, the axioms, and that the formal statement means what the prose says. The first two are shared with every Lean development. For the third, each result in the text names its declaration, and Section 2 describes the formal objects in enough detail that the statements can be read; the hypotheses in the text are the hypotheses in the code.

The household problem. The dynamic programme is built on a general stochastic dynamic-programming layer with extended-real rewards, so that $u = -\infty$ at zero consumption is handled without a consumption floor. The principle of optimality, existence and uniqueness of the bounded value function, Berge’s maximum theorem for the policy (which Mathlib lacks; ours is built on its hemicontinuity definitions), and the Blackwell contraction are theorems of that layer. The asset region $[a, \bar{a}]$ is a type parameter, which is what lets the same theorems serve Aiyagari and Huggett.

The rate as a state. Continuity of the policy in the interest rate is obtained by adjoining the rate to the state and solving one dynamic programme on the product, so that joint continuity in the state is continuity in the rate. This avoids any parametric-continuity

argument, which would need a modulus that the Bellman contraction does not obviously supply.

Secant slopes. Lipschitz bounds, monotonicity in assets, monotonicity in the rate and the Euler inequalities are all proved by comparing plans, with no derivative of the value function. Where the development proves the value function differentiable it uses the sandwich lemma of Clausen and Strub (2020), formalised as `DifferentiableSandwich`. The uniqueness proof does not depend on it.

Probability. Stationary distributions are Borel probability measures on a compact metric space, in Mathlib’s `ProbabilityMeasure` type with the weak topology; the Krylov–Bogolyubov existence argument, weak continuity of aggregation, and the Doeblin contraction on bounded continuous functions are proved in `Distribution/`. The normal distribution behind Tauchen’s method is Mathlib’s `gaussianReal`.

Reproducing the check. With Lean and Mathlib installed, `lake build` at the repository root compiles the development, and `#print axioms` on any declaration in the index reports the axioms it uses. The repository is public and the history records the order in which the results were obtained.